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Rock Piles as Kōura Habitat

Reinforcing kōura resilience using rock piles to enhance freshwater crayfish habitat
in an Aotearoa New Zealand lake

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1 Abstract

2 Placeholder — abstract to be written.

3 Introduction

4 Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater
5 fauna now threatened with extinction (Sayer *et al.*, 2025). Among the most endangered are
6 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
7 invasive species, and water quality decline (Richman *et al.*, 2015). As keystone species and
8 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and foodweb
9 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Momot,
10 1995; Collier, Parkyn & Rabeni, 1997; Parkyn, Collier & Hicks, 2001).

11 In Aotearoa New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura
12 (*Paranephrops planifrons* White, 1842) is a taonga (culturally treasured) species of significant
13 ecological and cultural importance to Māori, particularly for Te Arawa iwi of the Rotorua lakes
14 region in the North Island of Aotearoa (Kusabs & Quinn, 2009). Kōura support customary
15 fisheries using cultural practices such as tau kōura, yet populations in Lakes Rotoiti and
16 Rotorua have declined by up to 96 % over the past two decades, with consequent loss of
17 these customary practices (Kusabs *et al.*, 2015a; Kusabs *et al.*, 2026). The primary driver of
18 decline is predation by the invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819,
19 hereafter catfish), confirmed in these lakes in 2016 and 2018 (Francis, 2019; Grayling, 2020).
20 Catfish are native to eastern North America where they mostly live in muddy slow-moving
21 water bodies and can survive and spawn in a wide temperature range, tolerate poor water
22 quality, low oxygen concentration and have a broad diet (Scott & Crossman, 1973), making
23 them a highly adaptable/tolerant invader. Catfish were introduced in Aotearoa in 1877 as
24 sport fish (McDowall, 1994) with their distribution continuously spreading over the upper
25 North Island such as the Waikato River and in 1985 they were detected in Lake Taupō (Barnes
26 & Hicks, 2003). In Aotearoa catfish are a major predator on kōura, as 64 % of large catfish
27 (<250 mm fork length) had kōura in their stomach contents (Barnes & Hicks, 2003). In
28 combination with predation by catfish kōura are further compressed by multiple interacting
29 stressors that progressively restrict their available habitat. Kōura are naturally found in
30 complex rocky habitats (Jowett, Parkyn & Richardson, 2008; Kusabs, Quinn & Hamilton,
31 2015b) or in deeper parts of the lake to avoid predation during the day, migrating to the

1 productive littoral zone at night to forage (Deveich, 1979). However, thermal stratification
2 creates anoxic conditions in deeper waters, displacing kōura into the shallow littoral zone
3 (Kusabs *et al.*, 2026), precisely where catfish predation pressure is greatest, as catfish rarely
4 swim beyond 17 m depth (Dedual, 2002). This compression is further intensified by dense
5 invasive macrophyte beds obstructing kōura migration in the littoral zone (Kusabs & Quinn,
6 2009), which also intensify oxygen depletion in deeper waters by die-off events (Vincent, Gibbs
7 & Dryden, 1984; Hamilton, McBride & Uraoka, 2005). The importance of deep refuge is shown
8 by Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake
9 remains oxygenated below the thermocline (Verburg & Albert, 2019). In Lake Rotoiti this
10 deep refuge becomes anoxic and unavailable for kōura (Kusabs *et al.*, 2026). Next to direct
11 predation catfish exposure triggers anti-predator responses in kōura including increased refuge
12 use (Raven *et al.*, 2026a), reducing time available for foraging and potentially suppressing
13 growth and reproduction (Preisser, Bolnick & Benard, 2005; Sheriff *et al.*, 2020). Climate
14 change is expected to strengthen thermal stratification, expand anoxic zones (Woolway *et al.*,
15 2021), and increase the range of suitable catfish habitat in Aotearoa, further reducing kōura
16 persistence (Lee *et al.*, 2025).

17 The compression of kōura into the littoral zone makes the quality of nearshore habitats
18 increasingly important for kōura populations. In stream environments, kōura are positively
19 associated with structurally complex habitats including bank undercuts, woody debris, leaf
20 litter, and coarse substrates (Usio & Townsend, 2000; Parkyn & Collier, 2004; Jowett *et*
21 *al.*, 2008; Parkyn, Meleason & Davies-Colley, 2009). In lake environments benthic substrate
22 characteristics are a stronger predictor of kōura distribution than nutrient enrichment or
23 predator abundance (Kusabs *et al.*, 2015b), and kōura are positively associated with coarse
24 substrates and riparian vegetation (Raven *et al.*, 2026b). Coarse rocky substrate provides
25 refugia from predators and shelter during moulting (Streissl & Hödl, 2002). Experimental
26 studies confirm that shelter availability directly reduces catfish predation on crayfish (Adams,
27 2007; Zhao *et al.*, 2024), and kōura increase refuge use in the presence of catfish compared
28 to native predators (Raven *et al.*, 2026a), emphasising the importance of shelter availability
29 for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti,
30 where the majority of the shoreline is sandy with dense beds of introduced macrophytes and
31 little natural rocky structure. Adding rocky substrate to soft-bottomed lake areas has been
32 shown to significantly increase crayfish abundance (Johnsen & Taugbøl, 2008), and increasing
33 structural complexity in streams improves crayfish restocking success (Huolila, Marjomäki &
34 Laukkanen, 1997). Deploying habitat enhancement structures that add structurally complexity
35 in the littoral zone could potentially be a practical way to restore kōura populations in Lake

1 Rotoiti.

2 In this study habitat enhancement structures in the shape of stone piles were deployed along
3 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
4 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
5 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
6 kōura occupancy, abundance and biomass would increase at stone pile sites relative to control
7 sites after construction, and that catfish would not change. We further predicted that kōura
8 abundance would keep increasing progressively after construction, at least within the first 18
9 months, and that catfish would instead show seasonal patterns of shoreline use rather than a
10 construction-driven trend (Barnes, 1996; Dedual, 2002). Finally, we predicted that stone pile
11 sites would progressively converge toward kōura densities observed at natural rocky reference
12 sites, and that catfish would not show this pattern.

13 **Materials and Methods**

14 **Study location and construction**

15 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
16 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
17 stratifies for approximately nine months a year. Since 2008, nutrient-rich inflows from Lake
18 Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to
19 improve water quality in Lake Rotoiti (Hamilton *et al.*, 2005; Prentice & Özkundakci, 2024).
20 Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and
21 controlled water level throughout the year. Catfish are present throughout the lake but occur
22 in particularly high concentrations in Te Weta Bay on the North West side of the lake (Francis,
23 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and
24 land owner approval (Figure 1). At each stone pile site, a paired control site was established
25 50 m along the shoreline in the same habitat type. Five naturally occurring rocky shoreline
26 locations were selected as reference sites, each with two monitoring points 50 m apart. In total
27 20 sites of 150 m² (15 * 10 m) were monitored seasonally over 18 months. Baseline monitoring
28 was conducted in February 2025 prior to stone pile deployment, which occurred the following
29 week. Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and
30 Quarry, ~25 km from the lake) with a size range of 85 - 315 mm mean 173 mm. At each of the
31 five stone pile sites, three stone piles were arranged horizontally along the shoreline by hand

1 from the shore. Stone pile design was informed by mesocosm experiments demonstrating kōura
2 preference for enclosed stone structures (Raven *et al.*, in review). Stone piles were positioned
3 approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

4 **Monitoring**

5 Monitoring consisted of physical, chemical, and biological parameters recorded at each site
6 during each monitoring round. Static site characteristics including substrate composition, slope
7 to depth contour, riparian vegetation, overhanging tree cover and wood cover were visually
8 estimated at baseline monitoring only, as these were not expected to change substantially over
9 the monitoring period. Substrate composition was visually estimated using an underwater
10 viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64
11 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter. A substrate index was
12 calculated from these estimates (Jowett & Richardson, 1990; Jowett *et al.*, 2008):

$$\begin{aligned}\text{Substrate index} = & 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} \\ & + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} \\ & + 0.02 \times \text{organic matter} + 0.01 \times \text{mud}\end{aligned}$$

13 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
14 and stable structures used by kōura as refuge (Usio & Townsend, 2000; Nyström *et al.*, 2006).
15 After stone pile construction, measurements of the size and dimensions of each structure were
16 taken. The post-construction substrate index for stone pile sites was recalculated using each
17 site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
18 and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect
19 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
20 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
21 projected coordinate system (NZTM2000; EPSG:2193).

22 Seasonal variable parameters were recorded at every monitoring round, and included aquatic
23 vegetation cover (emergent and submerged macrophytes) and chemical parameters included
24 water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (μS cm⁻¹) were measured
25 using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured
26 using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

1 Kōura and fish were every monitoring round sampled using two single-winged, double-throated
 2 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm;
 3 two 50 mm entrance openings) baited with catfish flesh (MacNeil *et al.*, 2025). Fyke nets were
 4 placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals,
 5 with bait traps positioned between them. All equipment was deployed simultaneously at each
 6 site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using
 7 a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5
 8 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish
 9 species, including eels (*Anguilla* spp.), goldfish (*Carassius auratus* (Linnaeus, 1758)), common
 10 smelt (*Retropinna retropinna* (Richardson, 1848)), kōaro (*Galaxias brevipinnis* Günther, 1866),
 11 rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792), common bully (*Gobiomorphus cotidianus*
 12 McDowall & Fulton, 1978), and mosquitofish (*Gambusia affinis* (S. F. Baird & Girard, 1853)),
 13 were recorded for abundance and measured for total length, and weighed for wet mass. Catch
 14 per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and
 15 fish based on the number of nets and traps used at each site for each monitoring round (Kusabs
 16 *et al.*, 2015b). Missing kōura weights due to failure of scale (n = 36) were estimated using a
 17 log –log length–weight regression fitted to individuals with both measurements (n = 757),
 18 following standard approaches for weight length relationships in aquatic organisms ?@fig-
 19 **length-weight**. Predicted values were back-transformed with a lognormal bias correction
 20 factor ($10^{\{0.5 \hat{\sigma}^2\}}$) (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.683 \times \log_{10}(\text{OCL length [mm]})$$

21 Data analysis

22 Data analysis was divided into three sections, the first section assessed baseline equivalence
 23 between stone pile, control, and reference sites before construction, the second section assessed
 24 the impact of stone pile construction and after-construction trajectory on kōura and catfish
 25 occupancy, abundance, and biomass, and the third section assessed whether stone pile sites
 26 converged toward reference sites and if reference sites remained stable over time. All analysis
 27 was conducted in R (version 4.5.2; R Core Team (2025)) using the glmmTMB and emmeans
 28 packages (Brooks *et al.*, 2017; Lenth & Piaskowski, 2017).

29 Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank
 30 tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum

1 tests to compare stone pile sites with reference sites. This was done for response variables
2 (water quality, substrate, CPUE, BPUE) to confirm that the stone pile and control sites
3 were equivalent before construction, and to characterise the difference between stone pile and
4 reference sites. To further assess the baseline equivalence generalised linear models (GLMs) were
5 fitted testing the CPUE and BPUE of kōura and catfish against site type (stone pile/control)
6 and tested each of the two reference sites (reference 1/reference 2) with a Tweedie family and
7 log link. The covariate temperature was included in the catfish abundance and biomass models
8 testing the reference sites to allow model convergence.

9 The second section of the analysis tested construction impact relative to control sites using a
10 before-after, control-impact (BACI) design ([Underwood, 1994](#)) and tested after-construction
11 trajectory of kōura and catfish populations. The BACI design tested using generalised linear
12 mixed effects models (GLMMs) the effect of stone pile construction on kōura and catfish
13 occupancy, abundance, and biomass by comparing one pre-construction monitoring round with
14 six post-construction monitoring rounds. The models tested period (before/after) and site
15 type (stone pile/control) as an interaction term, with monitoring round as a covariate and
16 site nested within pair as a random effect. Occupancy was tested with a Binomial family and
17 abundance and biomass using a Tweedie family with log link. The after-construction trajectory
18 models were tested using GLMMs with site type (stone pile/control) and monitoring round
19 as interaction terms and site number as random effect using similar families as the BACI
20 models.

21 The third section of the analysis assessed whether stone pile sites converged toward reference
22 sites and if reference sites remained stable over time. Kōura and catfish abundance, and biomass
23 were tested in GLMMs fitted with site type (stone pile/reference) and monitoring round as
24 interaction terms with site number as random effect using a Tweedie family with log link. The
25 catfish abundance model also included temperature as a covariate. The reference site stability
26 was tested in GLMMs fitted with site type (reference 1/reference 2) and monitoring round as
27 interaction terms with site nested within pair as a random effect also using a Tweedie family
28 with log link. Temperature was included in the catfish abundance and season was included in
29 the kōura abundance as covariates.

30 Multiple-testing correction using Benjamini–Hochberg correction was applied within each
31 research sub-question across the set of models tested. Robustness of the models was further
32 assessed via sensitivity analyses and all models were tested for uniform distribution, dispersion,
33 and outlier rate using the DHARMA package ([Hartig, 2016](#)). Covariate selection for each model
34 was preformed by comparing Corrected Akaike Information Criterion (AICc) of the base

1 model against models with one additional covariate (temperature, season, or substrate index)
2 added at a time. A covariate was retained only when it reduced AICc by more than 2 units
3 relative to the base model (Burnham & R., 2002). Temperature was retained in in catfish
4 models where it substantially improved model fit, consistent with catfish being associated with
5 warmer water temperatures (Scott & Crossman, 1973). For the two reference-site catfish models
6 specifically, the base model failed to converge and temperature was required for numerical
7 stability. Season was retained in the kōura abundance model for reference stability, reflecting
8 the known seasonal variation in kōura abundance (Kusabs *et al.*, 2015b). Season could not be
9 evaluated for the BACI models as it is collinear with the before/after period contrast.

10 **Results**

11 Figure 2

12 Figure 4

13 *Baseline equivalence*

14 *Construction impact relative to control sites*

15 *Recovery toward reference conditions*

16 Figure 3

17 **Discussion**

18 *Opening*

19 **Baseline and site context***

20 **Construction impact***

21 **Convergence and recovery***

22 **Implications and future directions***

23 For future studies it is recommended

1 Most kōura were found along the edges of the stone piles there for recommended future stone
2 piles should be designed to create the most edges.

3 Stone piles were constructed at the shore line to shallow to reach macrophyte beds which
4 are located further offshore. Future stone pile construction could be extended further offshore
5 to reach the macrophyte beds and test if stone piles could suppress macrophyte growth and
6 create pathways for kōura to move between the deeperparts of the lake and the shore.

7 **Conclusion**

8 **Acknowledgements**

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22 **Authors’ contributions**

23 Study conception and design were undertaken by O.V.R., I.A.K.K., R.H., F.J.B. and D.Ö.
24 Fieldwork was executed by O.V.R. Data analysis was performed by O.V.R. I.A.K.K obtained
25 council and iwi consent and provided local ecological knowledge. Manuscript drafting was led
26 by O.V.R. All authors contributed to manuscript revision, approved the results, and the final
27 manuscript.

1 Use of Artificial Intelligence

2 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
3 language refinement. All content was critically reviewed and the authors take full responsibility
4 for the final manuscript.

5 Declarations

6 The corresponding author has declared that none of the authors has any competing interests.

7 Data availability statement

8 The code and derived data supporting the findings of this study are openly available on GitHub
9 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
10 is hosted at https://oliverraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
11 archived on Zenodo (<https://doi.org/.....>). Peer review documents, including the cover letter
12 and reviewer responses, are available in the manuscript repository in the interest of open and
13 transparent science.

14 Tables

Table 1: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test, pairs matched by pair_id, supporting the BACI design) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test, documenting the starting gap relative to natural conditions, motivating the convergence analysis). With only 5 paired sites, these are low-power checks — a non-significant result indicates no obvious gross imbalance, not formal proof of equivalence.

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205
Catfish CPUE	0.05 $\pm$ 0.11	0.1 $\pm$ 0.22	0.3 $\pm$ 0.52	1.000	0.420
Kōura BPUE	7.95 $\pm$ 12.15	2.8 $\pm$ 3.84	36.69 $\pm$ 36.82	0.789	0.206

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Catfish BPUE	0.15 ± 0.34	24.6 ± 55.01	97.28 ± 232.15	1.000	0.343
Substrate index (before)	3.39 ± 1.07	3.68 ± 0.95	5.29 ± 1.05	0.789	0.012
Substrate index (after)	3.65 ± 1.12	3.68 ± 0.95	5.29 ± 1.05	1.000	0.032
Slope (0–20 m)	0.06 ± 0.05	0.06 ± 0.04	0.09 ± 0.05	0.590	0.245
Temperature (°C)	24.74 ± 1.92	24.78 ± 1.47	24.48 ± 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 ± 1.48	9.63 ± 1.29	10.03 ± 0.86	0.418	0.426
Specific conductivity (µS/cm)	195.44 ± 70.05	199.8 ± 78.52	165.45 ± 8.5	0.281	0.951
pH	7.5 ± 2.15	7.61 ± 2.2	8.53 ± 0.66	1.000	0.426

¹ Source: [Article Notebook](#)

² Figures

³ Supplementary information

⁴ *Supplementary figures are provided in the appendices.*

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¹⁵ *tributed Packages*



Figure 1: Map of the monitoring sites.

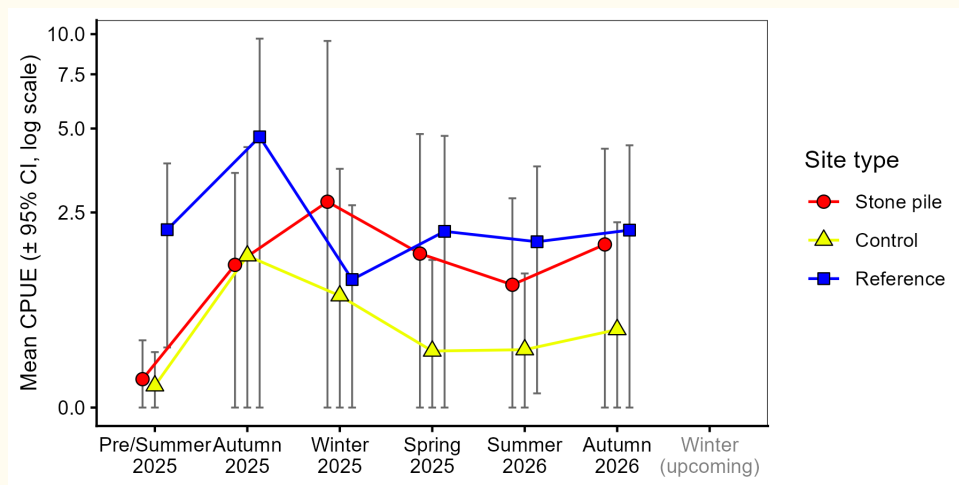


Figure 2: Mean catch per monitoring round ($\pm$ 95% CI) by site type for kōura.

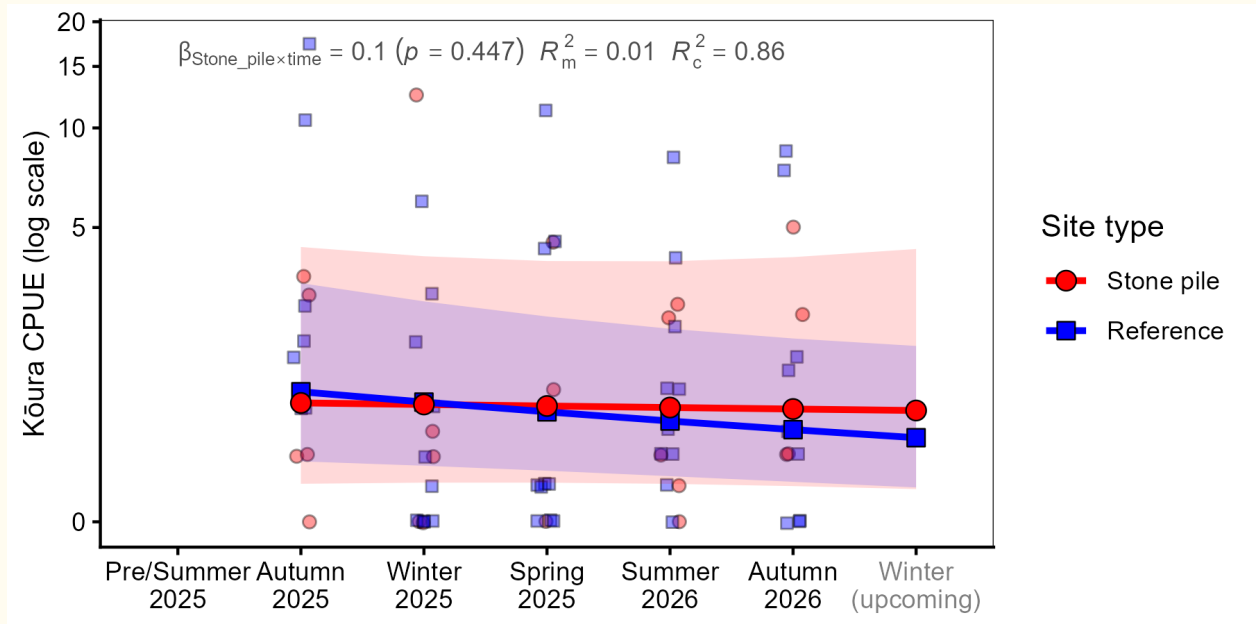


Figure 3: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the RQ3 abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\beta \pm p\text{-value}$) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

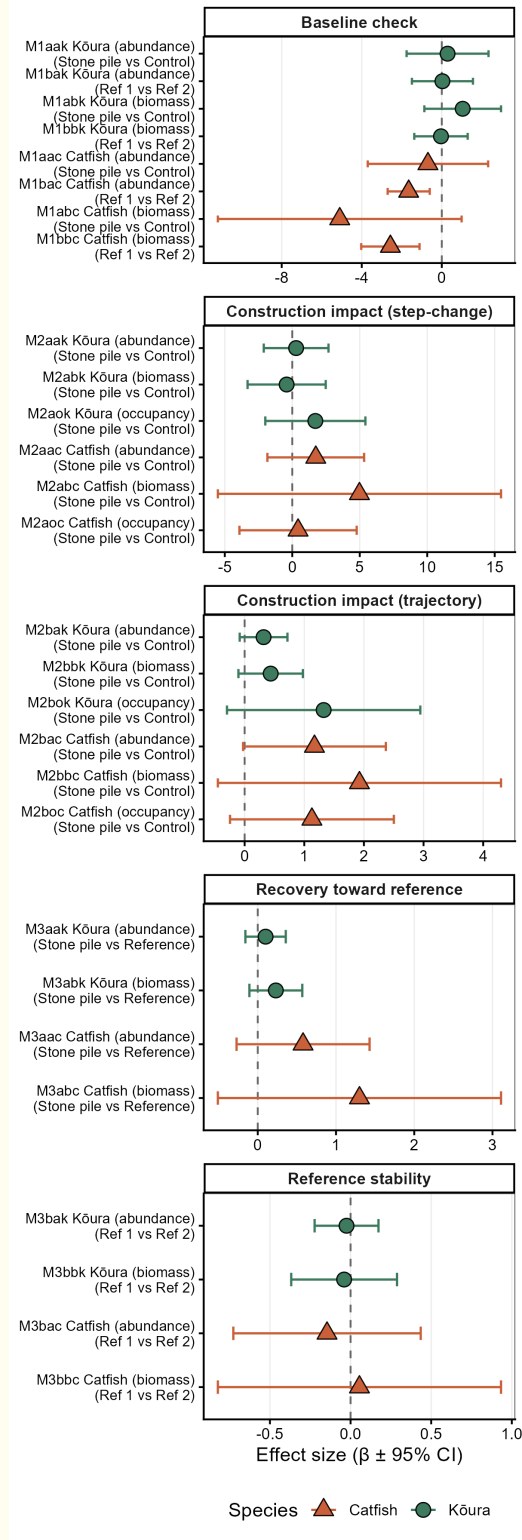


Figure 4: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest of every model in model_registry (kōura and catfish, RQ1–RQ4). Filled points indicate significant effects ($p < 0.05$). Each panel has its own x-axis scale— β magnitudes are not directly comparable across panels because each RQ tests a different contrast (step-change, per-round slope, or simple group difference) and RQ1 occupancy is on the log-odds scale while all others use the Tweedie log link. Models with degenerate confidence intervals (non-positive-definite Hessian) are excluded.

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